

# ACM TKMCE WEBSITE

## SOFTWARE DESIGN DOCUMENT

### 1. INTRODUCTION

#### 1.1 Purpose

The purpose of this Software Design Document (SDD) is to define the design and overall structure of the ACM TKMCE Website. It serves as a reference for the development team by describing the website's layout, functionality, user interactions, and key system components before the implementation phase.

This document aims to provide a common understanding among all team members, ensuring consistency throughout the design and development process while simplifying future maintenance and enhancements.

#### 1.2 Scope

The ACM TKMCE Website is intended to serve as the official online platform for the ACM Student Chapter at campus. The website will provide information about the chapter, its vision, mission, executive committee, events, testimonials, and contact details while also supporting event registration and management.

The system will enable users to explore upcoming events, register through the website, and securely store registration details in a centralized database. It will also provide administrative features for managing events, registrations, and chapter content.

Features such as advanced analytics, mobile applications, and integrations with external platforms are considered outside the scope of the initial version.



## 1.3 Definitions and Acronyms

| Term                       | Definition                                                                    |
|----------------------------|-------------------------------------------------------------------------------|
| Hero Section               | Introductory section displayed when a user first accesses the website.        |
| CTA(Call-to-Action) Button | User interface element that encourages visitors to perform a specific action. |
| TBF                        | To be finalized                                                               |



## 1.4 References

The following resources were referred to during the planning and design of the ACM TKMCE Website:

1. ACM Official Website – <https://www.acm.org/>
2. ACM Student Chapters–<https://www.acm.org/chapters/student-chapters>

## 2. SYSTEM OVERVIEW

### 2.1 Project Description

The ACM TKMCE Website provides information about the chapter, showcases its activities and events, and offers a centralized system for event registration and management.

### 2.2 Objectives

- To provide a centralized platform for ACM TKMCE.
- To showcase the chapter's activities, events, and achievements.
- To simplify event registration through an online system.
- To provide administrators with tools for event management.

## 2.3 User Roles

| User Role              | Description                                                                                                       |
|------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Visitor</b>         | Can browse the website and view publicly available information such as events, chapter details, and testimonials. |
| <b>Registered User</b> | Can log in, register for events, and access features available to authenticated users.                            |
| <b>ACM Member</b>      | A registered user with ACM membership who may receive additional benefits such as discounted event registrations. |
| <b>Administrator</b>   | Manages website content, events, registrations, and user information.                                             |

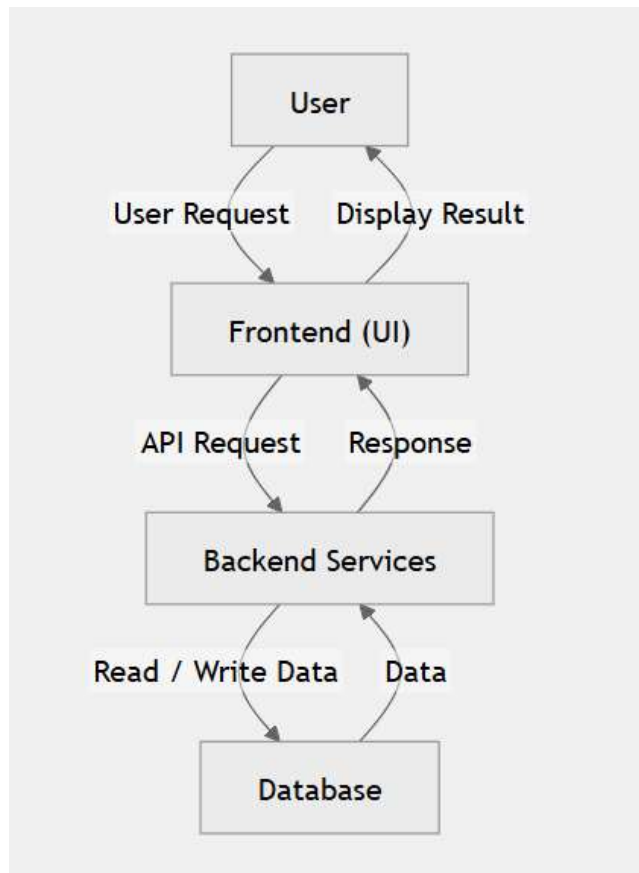
## 3. Architectural Design

### 3.1 Technology Stack

|                  |              |
|------------------|--------------|
| Backend:         | TBF          |
| Database:        | TBF          |
| Authentication:  | TBF          |
| Deployment:      | TBF          |
| Version Control: | Git & GitHub |

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### 3.4 Data Flow and Control Flow



## 4.Screen Design

### 4.1 HOME PAGE

#### Purpose

The Home Page serves as the primary entry point to the ACM TKMCE website. It provides visitors with an overview of the chapter, its goals, leadership, activities, achievements, and contact information while enabling easy navigation to other sections of the platform.

# Components

## 1. Hero Section

The Hero Section shall be the first section displayed when a user opens the website. It should provide an engaging introduction to the ACM TKMCE Student Chapter and create a strong first impression.

### Contents

- ACM Student Chapter Logo
- Chapter Name (ACM TKMCE)
- Short introductory description of the chapter
- Optional Call-to-Action Button
  - Join ACM
  - Explore Events
  - Learn More

The design should be visually appealing and welcoming, similar to a modern landing page.

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## 2. Navigation Bar

The navigation bar shall remain visible at the top of the page, allowing users to navigate easily between different sections of the website.

### Menu Options

- About
- Goals
- Execom
- Events
- Testimonials
- Contact

## 1.About

- About ACM
  - About ACM TKMCE
  - Year of Establishment
  - Chapter Statistics (Number of Members,Events Conducted,Workshops,Achievements)
  - Why Join ACM? (Display as Cards)
- 



## 2.Goals

- Vision
  - Mission
  - Values
- (Display as cards)
- 

## 3.Execom

**Option 1:** Team-wise group photos

**Option 2:** Individual member cards under each team

- Photo
  - Name
  - Position
  - LinkedIn
  - GitHub
- 

## 4.Events

- Upcoming Events
- Event Cards
- Register Button
- View Details

## Testimonials

- Carousel Layout
  - Alumni Photo
  - Current Position
  - Company
  - Testimonial
- 



## Contact

- "Let's Connect"
- LinkedIn
- GitHub
- Email

**Optional:** Contact Form *(To be finalized)*

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## Optional Sections

- Gallery Preview
- Updates & Announcements

## 3. Social Media Links

- Instagram
- LinkedIn
- GitHub

The social media icons shall be displayed on all pages and redirect users to the official ACM TKMCE social media profiles.

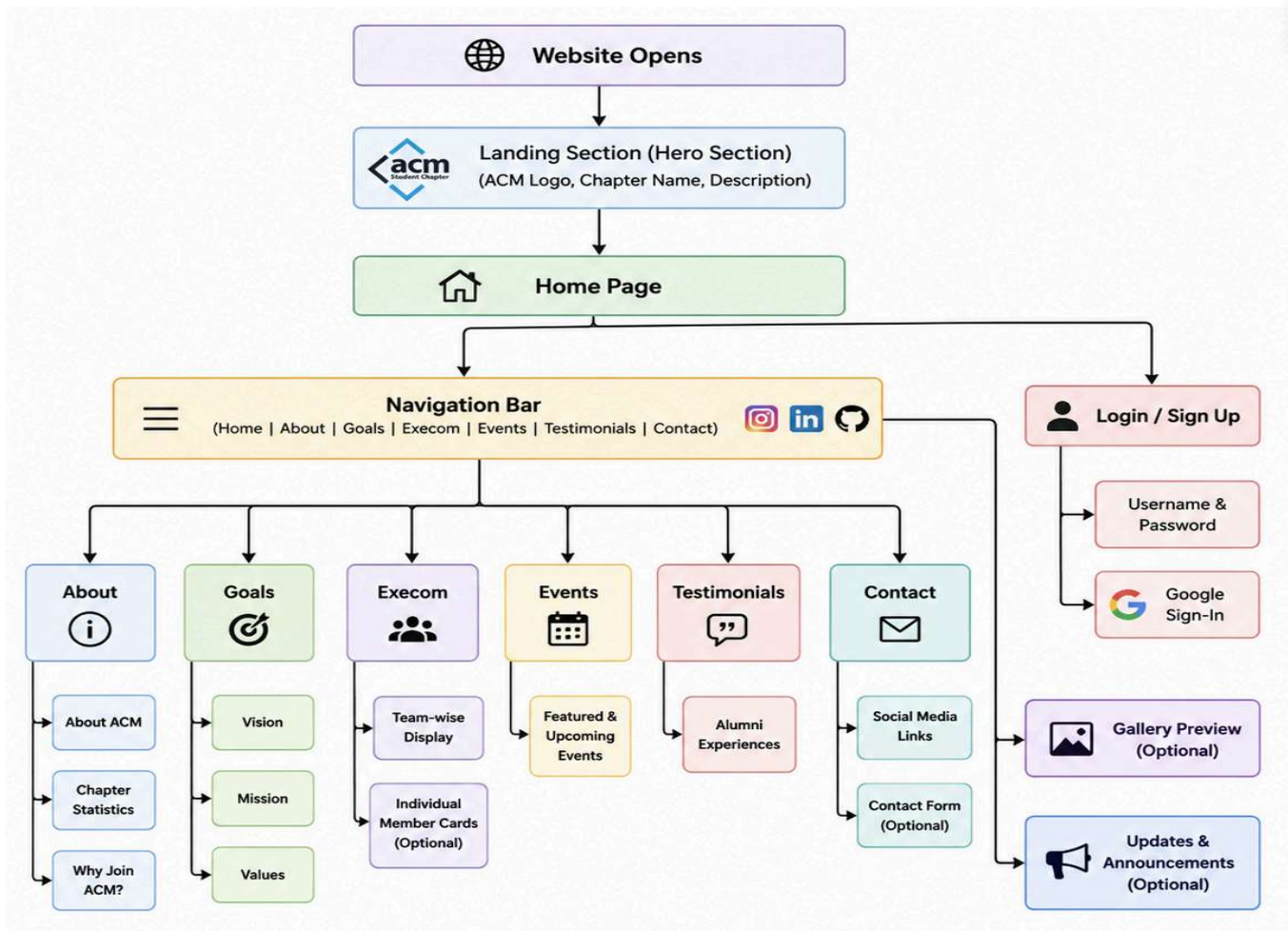
## 4. Authentication Access

A dedicated **Login / Sign Up** component shall be provided separately from the navigation bar.

## Supported Authentication Methods

- Username and Password
- Google Sign-In

The final placement and user interface of this component shall be decided during the UI design phase.



## 4.2 AUTHENTICATION PAGE

### Purpose

The Login / Sign Up page enables users to securely access the ACM TKMCE Website and Event Management Portal. Authenticated users can register for events, access personalized features, and manage their profile.



## Login / Sign Up Flow

Home Page



Login / Sign Up



Google Authentication



Google Account Verification



User Dashboard / Previous Page



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## Components

### 1. Google Authentication

Google Authentication shall serve as the primary login and registration method. Users can securely sign in using their Google account without creating a separate password. On successful authentication, the system retrieves the necessary user information and grants access to the website.

### 2. Optional Password Authentication

The system may optionally support authentication using an email/username and password. This feature may be implemented in future versions based on project requirements.

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## User Actions

- Sign in using a Google account.
- Create an account automatically during the first Google Sign-In.
- Log out of the system.

## Navigation

### Accessible From

- Home Page
- Event Registration Page

### Navigates To

- User Dashboard
- Event Registration Page (if redirected after login)
- Previous Page



### Notes

- Google Authentication is the preferred authentication method for improved user convenience and security.
- Password-based authentication is considered an optional feature and may be incorporated in future versions.

## 4.3 EVENTS PAGE

### Purpose

The Events Page provides users with a centralized platform to browse all ACM TKMCE events. It displays upcoming, ongoing, and past events in an organized manner, allowing users to explore available events and access detailed information.

### Events Page Flow

Home Page

|

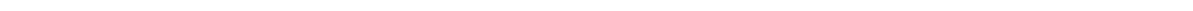
Events Page

|

Event Cards

|

Event Details Page



## Components

### 1. Event Cards

The Events Page primarily consists of **Event Cards**, each representing an individual ACM event.

Each Event Card may display:

- Event Poster
- Event Name
- Date & Time
- Venue
- Short Description
- Registration Status (Open / Closed)

Clicking anywhere on an Event Card redirects the user to the corresponding **Event Details Page**, where complete information about the selected event is displayed.



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## User Actions

A user can:

- Browse available events.
- Select an event by clicking on its Event Card.
- Navigate to the Event Details Page.

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## Navigation

### Accessible From

- Home Page (Navigation Bar)

### Navigates To

- Event Details Page

## Future Enhancements

The following features may be incorporated in future versions:

- Search Bar
- Event Filters
- Featured Events
- Event Sorting (Date, Category, Registration Status)



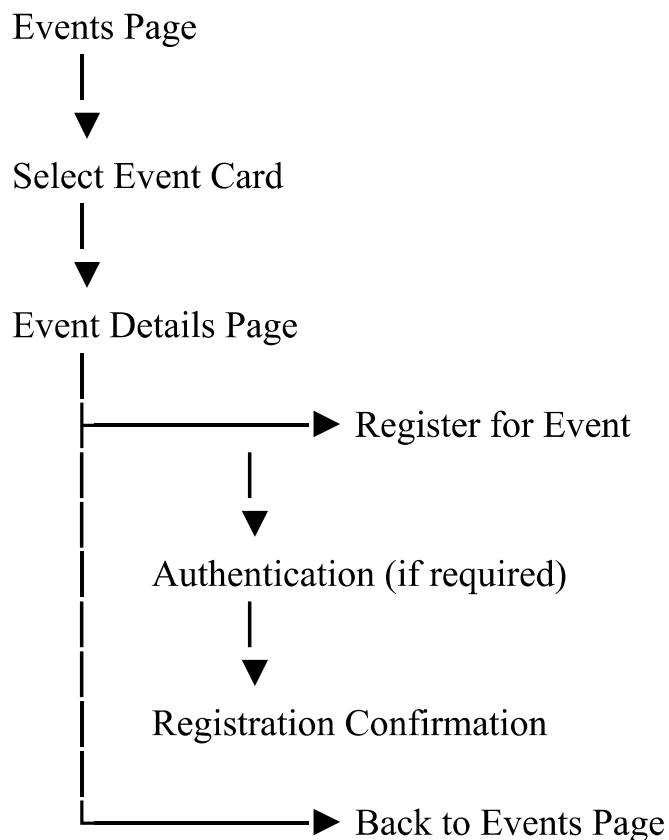
## 4.4 EVENT DETAILS PAGE

### Purpose

The Event Details Page provides comprehensive information about a selected ACM event. It enables users to understand the event, view important details, and register for participation through the website.

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### Event Details Page Flow



# Components

## 1. Event Header

The Event Header serves as the primary information section of the page.

It shall include:

- Event Banner / Poster
- Event Name
- Date
- Time
- Venue

A **Register Now** button shall be positioned near the **top-right section of the Event Header**, ensuring that it remains easily accessible without requiring users to scroll through the page.



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## 2. Event Description

Displays a detailed overview of the event, including its objectives, highlights, and other relevant information.

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## 3. Speaker / Mentor Details (*Optional*)

Displays information about the speakers, mentors, or guests associated with the event.

The section may include:

- Name
  - Designation
  - Organization
  - Photograph (*Optional*)
-

## 4. Registration Information

Displays important registration details, including:

- Registration Status
- Registration Deadline (*if applicable*)
- Registration Fee (*if applicable*)

Registration Status may be represented using status badges such as:

-  Registration Open
-  Registration Closed
-  Coming Soon



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### User Actions

A user can:

- View complete event information.
- Register for the event.
- View the event poster in full-screen mode (*optional*).
- Return to the Events Page.

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### Navigation

#### Accessible From

- Events Page

#### Navigates To

- Authentication Page (*if login is required*)
  - Registration Confirmation Page
  - Events Page
-

## Future Enhancements

The following features may be incorporated in future versions:

- Event Agenda / Schedule
- Seat Availability
- Add to Google Calendar
- Share Event

## Alternative Design Consideration

As an alternative to a dedicated **Event Registration Page**, the registration process may be implemented as a **modal (popup) window** that appears when the user clicks the **Register Now** button.

This approach enables users to complete the registration process without leaving the Event Details Page

## 4.5 Event Registration Page

### Purpose

The Event Registration Page enables users to register for a selected ACM event by providing the required information. It collects participant details, verifies eligibility where applicable, and completes the registration process.

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### Event Registration Flow

Event Details Page

|

Click Register Now

|

Authentication (if required)

|

Event Registration Page

|

Submit Registration

|

Registration Confirmation

---



## Components

### 1. Event Summary

Displays a brief summary of the selected event.

It shall include:

- Event Name
  - Date
  - Time
  - Venue
- 

### 2. Participant Information

Collects the participant's details.

Fields may include:

- Full Name
- Email Address
- Phone Number
- College
- Department
- Year of Study



### 3. ACM Membership Verification (*Optional*)

If member benefits are available, the system may request:

- ACM Membership ID

The system verifies the membership and applies the corresponding benefits, such as discounted registration fees.

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### 4. Payment Section (*If Applicable*)

Displays:

- Registration Fee
- Payment Method
- Payment Status

If the event is free, this section may be omitted.

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### 5. Terms and Conditions

Users must agree to the event guidelines before submitting their registration.

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### 6. Submit Registration Button

Completes the registration process and redirects the user to the Registration Confirmation Page.

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### User Actions

A user can:

- Enter registration details.

- Verify ACM membership (if applicable).
  - Complete payment (if applicable).
  - Submit the registration.
  - Cancel and return to the Event Details Page.
- 

## Navigation

### Accessible From

- Event Details Page

### Navigates To

- Registration Confirmation Page
  - Event Details Page
- 

## Future Enhancements

The following features may be incorporated in future versions:

- Team Registration
- Digital Event Pass
- Automatic Email Confirmation

## 4.6 User Dashboard

### Purpose

The User Dashboard provides registered users with a personalized space to manage their account and access event-related information. It enables users to view their profile, track registered events, and access features available to authenticated users.

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## User Dashboard Flow

Login Successful



User Dashboard



View Profile



Registered Events



Browse Events



Logout



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## Components

### 1. User Profile

Displays the basic details of the logged-in user.

It may include:

- Profile Picture
  - Name
  - Email Address
  - Department
  - Year of Study
  - ACM Membership Status (*if applicable*)
-

## 2. Registered Events

Displays all events for which the user has successfully registered.

Each registered event may display:

- Event Name
- Date
- Venue
- Registration Status

Clicking on an event redirects the user to the corresponding **Event Details Page**.

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## 3. Browse Events

Provides quick access to the Events Page, allowing users to explore and register for additional events.

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## 4. Account Settings (*Optional*)

Allows users to update basic profile information.

This feature may be expanded in future versions.

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## 5. Logout

Allows users to securely end their current session and return to the Home Page.

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## User Actions

A user can:



- View profile information.
  - View registered events.
  - Browse available events.
  - Update profile information (*optional*).
  - Log out of the system.
- 

## Navigation

### Accessible From

- Authentication Page
- Event Registration (after successful login)

### Navigates To

- Events Page
  - Event Details Page
  - Home Page (after logout)
- 

## Future Enhancements

The following features may be incorporated in future versions:

- Event Certificates
- Notification
- Edit Profile

## 4.7 Admin Dashboard

### Purpose

The Admin Dashboard provides authorized administrators with a centralized interface to manage the ACM TKMCE Website. It enables administrators to create and manage events, monitor registrations, update website content, and perform other administrative tasks.

# Admin Dashboard Flow

Admin Login



Admin Dashboard



Manage Events



View Registrations



Manage Website Content



Manage Users



Logout



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## Components

### 1. Dashboard Overview

Displays a summary of website statistics, such as:

- Total Events
  - Total Registrations
  - Active Users (*Optional*)
  - Upcoming Events
-

## 2. Event Management

Allows administrators to:

- Create Events
  - Edit Event Details
  - Delete Events
  - Update Registration Status
- 



## 3. Registration Management

Displays registrations for each event.

Administrators can:

- View registered participants
  - Verify ACM member registrations (*if applicable*)
  - Export participant lists (*Optional*)
- 

## 4. Website Content Management

Allows administrators to manage website content, including:

- Home Page Content
  - Testimonials
  - Execom Members
  - Gallery
  - Announcements
- 

## 5. User Management (*Optional*)

Allows administrators to:

- View registered users
- Manage user information

- Assign administrative roles (*Future*)
- 

## 6. Logout

Allows administrators to securely log out of the system.

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### User Actions

An administrator can:

- Create, edit, and delete events.
  - View and manage event registrations.
  - Update website content.
  - View registered users.
  - Log out of the system.
- 

### Navigation

#### Accessible From

- Authentication Page (*Admin Login*)

#### Navigates To

- Event Management
  - Registration Management
  - Home Page (after logout)
- 





## 5. DATABASE DESIGN

### 5.1 Purpose

The database is designed to store and manage the information required for the ACM TKMCE Website and Event Management Portal. It maintains records related to users, events, registrations, executive committee members, testimonials, and gallery content while ensuring efficient storage and retrieval of data.



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### 5.2 Database Entities

#### 1. User

Stores information about registered users of the website.

The entity may include:

- User ID
- Name
- Email Address
- Phone Number
- Department
- Year of Study
- ACM Membership Status

---

#### 2. Event

Stores information related to ACM events.

The entity may include:

- Event ID
- Event Name
- Description

- Date
  - Time
  - Venue
  - Registration Deadline
  - Registration Fee
  - Registration Status
  - Event Banner / Poster
- 



### **3. Registration**

Stores details of users registered for events.

The entity may include:

- Registration ID
  - User ID
  - Event ID
  - Registration Date
  - Registration Status
  - Payment Status (*if applicable*)
- 

### **4. Executive Committee (Execom)**

Stores information about the ACM Executive Committee members.

The entity may include:

- Member ID
  - Name
  - Position
  - Team
  - Photograph
  - LinkedIn Profile
  - GitHub Profile
-

## 5. Testimonials

Stores testimonials displayed on the website.

The entity may include:

- Testimonial ID
- Name
- Designation
- Organization
- Testimonial Content
- Photograph (*Optional*)



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## 6. Gallery

Stores images related to ACM events and activities.

The entity may include:

- Image ID
- Event ID (*Optional*)
- Image

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## 5.3 Entity Relationships

The primary relationships between the database entities are as follows:

- A **User** can register for multiple **Events**.
  - An **Event** can have multiple **Registrations**.
  - Each **Registration** is associated with one **User** and one **Event**.
  - An **Event** may contain multiple **Gallery** images.
  - **Executive Committee** and **Testimonials** are independent entities managed by the administrators.
-

## 5.5 Future Considerations

The database design may be extended in future versions to support additional features, including:

- Certificate Management
- Notifications
- Event Analytics



## 6. FUNCTIONAL REQUIREMENTS

### 6.1 User Authentication

The system shall:

- Allow users to sign in using Google Authentication.
  - Authenticate users before accessing restricted features.
  - Allow users to securely log out of the system.
- 

### 6.2 Event Management

The system shall:

- Display all available events.
  - Allow users to view complete event details.
  - Display registration status for each event.
  - Allow administrators to create, update, and delete events.
- 

### 6.3 Event Registration

The system shall:

- Allow authenticated users to register for events.
- Collect participant information during registration.
- Verify ACM membership where applicable.
- Display a confirmation message upon successful registration.

## 6.4 User Dashboard

The system shall:

- Display the user's profile information.
  - Display registered events.
  - Allow users to browse additional events.
  - Allow users to log out securely.
- 



## 6.5 Admin Dashboard

The system shall:

- Allow administrators to manage events.
  - View event registrations.
  - Update website content.
  - Manage executive committee information.
  - Manage testimonials and gallery content.
- 

## 6.6 Website Content Management

The system shall:

- Display information about ACM TKMCE.
  - Display the chapter's Vision, Mission, and Values.
  - Display executive committee members.
  - Display testimonials.
  - Display contact information.
- 

## 6.7 Navigation

The system shall provide navigation between:

- Home Page

- Authentication Page
- Events Page
- Event Details Page
- Event Registration Page
- User Dashboard
- Admin Dashboard

## 7. Non-Functional Requirements

**Performance:** The system should provide quick response times during page navigation, event browsing, and event registration.

**Security:** The system should ensure secure authentication and protect user information from unauthorized access.

**Reliability:** The system should operate consistently under normal usage conditions and minimize failures during user interactions.

**Usability:** The website should provide a simple, intuitive, and user-friendly interface for all users.

**Scalability:** The system should support future enhancements and accommodate increased user traffic without significant redesign.

**Maintainability:** The system should be modular and well-structured, enabling easy updates and maintenance.

**Availability:** The website should remain accessible whenever required, except during scheduled maintenance or unforeseen technical issues.

